Sahaj Bhadja

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Summary

Backend engineer with 2+ years building distributed services, real-time workflows, and AI-backed platforms. Strong in Python, SQL, cloud infrastructure, CI/CD, and reliable alerting and orchestration systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected **multi-tenant** AI backend services on **Azure SQL**, **Blob Storage**, and relational schemas for 500+ users, optimizing indexed queries and access patterns to reduce dashboard latency from 1.2s to 350ms.
- Built fault-tolerant **async workflows** across **FastAPI** background workers, processing 1K+ AI jobs with explicit state transitions and failure isolation, increasing throughput 3x under concurrent load while preserving reliable execution.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered **Flask** backend services and **REST APIs** for AI grading workflows serving 10K+ daily requests, integrating secure file ingestion, scoring pipelines, and instructor notifications across customer-facing production systems.
- Designed concurrent batch-processing on **AWS RDS** with multithreading and fault-tolerant execution, processing 1K+ file batches while maintaining stable API latency and reducing turnaround time by 60% during peak usage.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built spatial-temporal planning pipelines over 100K+ trajectory logs, integrating map geometry and agent context to improve closed-loop decision stability by 15% in safety-critical autonomous-driving simulations with sensor-driven feedback loops.
- Developed evaluation workflows for simulation-heavy ML systems, strengthening reproducibility, debugging, and failure analysis while collaborating on real-time model behavior across sensor-driven, latency-sensitive, high-iteration experimental environments and workloads.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **microservice**-based data-sharing software with **Jenkins**, **PyTest**, and **Mockito** automation across Windows, Linux, and macOS, improving release reliability through 20+ unit and integration test suites in production.
- Optimized large-scale **API** and **ETL** workflows with **AWS Glue**, **S3**, and object-oriented design patterns, improving response times by 30% and increasing pipeline efficiency by 68% under production constraints.

Skills

Languages: Python, JavaScript, TypeScript, SQL, Java, C++, Bash
Backend & Distributed Systems: FastAPI, Flask, Node.js, REST APIs, WebSockets, microservices, background workers, distributed systems, event-driven workflows, fault tolerance
Databases & Data: PostgreSQL, MySQL, MongoDB, SQL Server, Azure SQL, Redis familiarity, query optimization, schema design, ETL pipelines, data validation
Cloud & Infrastructure: AWS, Azure, GCP, Docker, Kubernetes, AWS RDS, S3, Blob Storage, Jenkins, CI/CD, Linux
AI & Engineering: multimodal pipelines, OCR workflows, real-time notifications, unit testing, integration testing, code reviews, Agile/Scrum, debugging

Projects

HMIS Live | *MERN, TypeScript, ABDM APIs, Twilio, Sentry*
- Implemented consent-based patient-history access, **Twilio** appointment alerts, **Sentry** monitoring, and scalable TypeScript-backed workflows to reduce administrative workload by 30%, emphasizing reliable event-driven notifications and operational visibility across systems.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, Plaid*
- Built a scalable backend supporting 50,000+ users and 1M+ transactions, combining real-time synchronization, secure token workflows, and deployment automation to maintain reliable high-volume APIs under continuous activity.

CPU-scheduling Simulator Live | *C++, React, EJ-2*
- Implemented FCFS, SJF, SRT, RR, priority scheduling, and MLQ with C++ logic, dynamic metrics, and Gantt visualization across 1K+ simulated process combinations for repeatable systems experimentation and latency reasoning.